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Easily construct, covert or build Likert items from vectors or data frames

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Executive Summary

This proposal outlines the development of open-source R package for converting or constructing Likert items from vectors or data frames, particularly from non-standard data structures. This proposal eliminates fragmented, manual workarounds for non-standard Likert-type data by introducing a reproducible workflow that mitigates data manipulation risks. The project is for a nine-month timeline and a budget of USD8,500. The timeline is divided into two-month inception phase to design the architecture, a four-month coding and testing phase to release the core engine, a two-month preliminary release phase for initial CRAN submission and documentation setup, and a final month for project closure. The budget directly funds labor at USD50 per hour, documentation hosting, and a sensitization workshop for faculty and students at the Federal University of Technology, Minna. Upon completion, the package and its continuous updates will be maintained publicly in GitHub and CRAN repositories.

Signatories

Project team

Professor Job Nmadu is leading the task and would deliver on the targets.

Contributors

Dr. Halima Sallawu of Federal University of Technology, Minna, Nigeria and I have participated actively in several data collection which necessitated this project.

Consulted

The Problem

The CRAN packages for Likert-type data workflow, viz, Mollazehi (2025); Winzar (2026) and Bryer and Speerschneider (2025), did not encompass transliteration to other formats for more robust interpretation, but, facilitates standard Likert items workflows. They work on conventional, clean ordinal format lacking the architectural framework for transliteration.

Empirical workflow may encounter non-standard data requiring transformation, like converting Likert-type items into ordinal scales and vice versa, or synthesizing valid Likert-type scales from non-Likert data. The existing workflow cannot execute these transformations natively, therefore, efforts has to rely on disjointed, manual workarounds and ad-hoc scripting. This fragmentation disrupts analysis, introduces data entry risks and errors, resulting in man-hours' loss. This project proposes an open-source R package for transliteration and dynamic format conversion to fill the gap while preserving data integrity, and handling non-standard Likert-type natively.

The proposal

Provide algorithms for the following concepts: binary construction, likert construction, likert conversion, vector description, and weight construction

Overview

The proposal is to create R package for Likert-type data workflows.

Detail

Minimum Viable Product

The package, focusing on Likert-type data structures for construction and conversion is hosted on GitHub.

Architecture

The package will be built in R statistical computing ecosystem with minimal external dependencies. The internal logic will revolve around transliteration and dynamic conversion algorithms.

Assumptions

There is high expectation that the project would be delivered with no particular risk associated with all timelines, except if emergencies or natural disasters, like COVID-19 happens.

External dependencies

The proposed package will rely of `tidyverse` established routines and workflows.

Project plan

Start-up phase

The start-up of the open-source R package, under GPL licence, is aimed at establishing the GitHub repository and integrating it with local machine workflows. The overall system architecture design

and defining the initial abstract classes that will govern the package's functionality would be the priority.

Technical delivery

Functions and codes in GitHub and CRAN repositories as well as continuous development and integration on feedback, stakeholder concerns and updates occasioned by change of rules by R Core teams.

Other aspects

Announcement posts and delivery blogs shall be used for continuous feedback while social media posts will be used for stakeholder engagement.

Budget & funding plan

Milestone	Period (months)	Deliverable/Tangible Products
Inception and take-off	2	Complete architecture design and initial abstract class definitions in the GitHub repository.
Coding and testing	4	Release the optimized core chunking engine; pass integration tests with local memory backends. Conduct a sensitization workshop for faculty and students.
Preliminary release and initial CRAN submission	2	Comprehensive pkgdown documentation, website hosting, initial CRAN submission, and host a community webinar.
Final release and closure	1	Final CRAN submission and project closure

In line with the above timelines, the budget propped for this project is USD8500 appropriated as follows:

Inception and take-off (USD1500): Funds 30 hours of core development work at a rate of USD50/hour covering GitHub setup and integration with local machine codes.

Coding and testing (USD5000): Funds 100 hours of setting up functions and algorithms, and documentation (website and GitHub repositories). Initial submission to CRAN.

Sensitization workshop(USD1000): Sensitization workshop and stakeholder engagement for faculty and students at Federal University of Technology, Minna. Publicity over social media handles.

Final release and closure (USD1000): Funds 20 hours for incorporating views from stakeholders into the the package, responding to CRAN reviews and comments, cleaning up the package and final release to CRAN.

Success

Definition of done

Measuring success

- Creation of the project in R and integration with GitHub
- Functions for the tasks created
- Testing the functions and sharing results
- Submission to CRAN

Future work

Bryer, Jason, and Kimberly Speerschneider. 2025. *Likert: Analysis and Visualization Likert Items*.
<https://doi.org/10.32614/CRAN.package.likert>.

Mollazehi, Mohammad. 2025. *LikertEZ: Easy Analysis and Visualization of Likert Scale Data*.
<https://doi.org/10.32614/CRAN.package.LikertEZ>.

Winzar, Hume. 2026. *LikertMakeR: Synthesise and Correlate Likert Scale and Rating-Scale Data Based on Summary Statistics*. <https://doi.org/10.32614/CRAN.package.LikertMakeR>.